

Creating a Dynamic Navigation Bar using SPFx Application Customizer

Building a Dynamic Navigation Bar using SPFx Application Customizer consists of several tasks starting from how to set your development environment and building the navigation component itself, deploying it to a SharePoint Online site, and configuring the dynamic navigation links. In this scenario, a detailed explanation on how to implement such a project using SPFx(Sharepoint Framework) and the Application Customizer extension is provided.

Technology Used

SharePoint Framework (SPFx)

React

PnPjs

Fluent UI

Table of Content:

Project Setup

Creating the Application Customizer

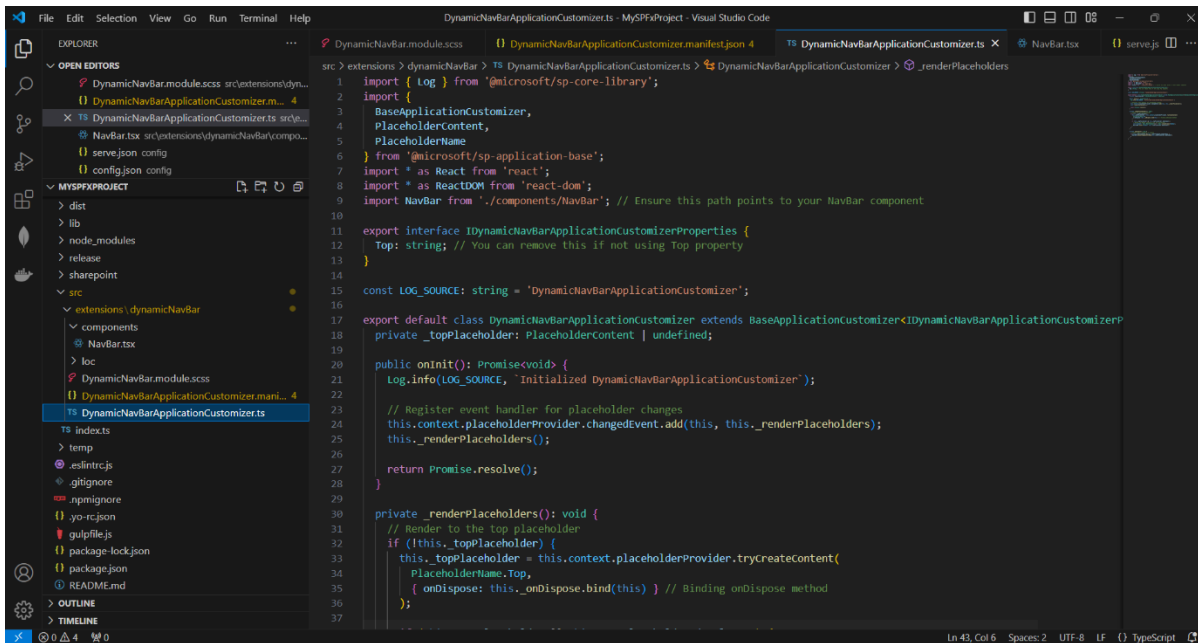
Building the Dynamic Navigation Bar

Renders the Navigation Bar

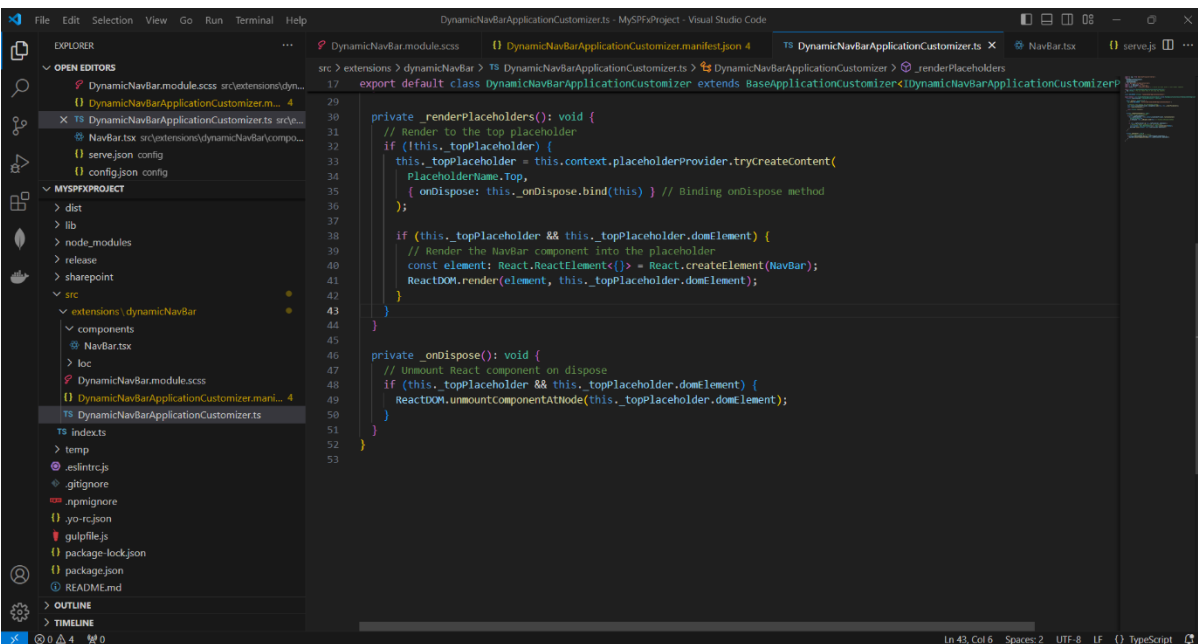
Deploying to SharePoint Online

Configuring Dynamic Navigation

Application Customizer(.ts) file

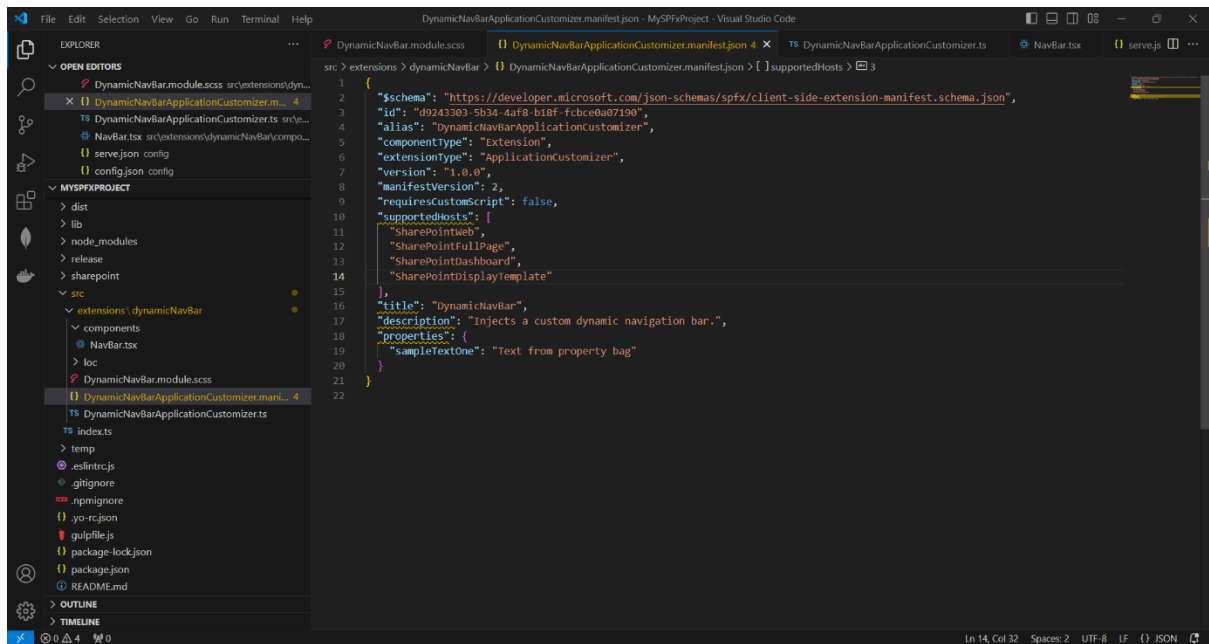


```
1 import { Log } from '@microsoft/sp-core-library';
2 import {
3   BaseApplicationCustomizer,
4   PlaceholderContent,
5   PlaceholderName,
6 } from '@microsoft/sp-application-base';
7 import * as React from 'react';
8 import * as ReactDOM from 'react-dom';
9 import NavBar from './components/NavBar'; // Ensure this path points to your NavBar component
10
11 export interface IDynamicNavBarApplicationCustomizerProperties {
12   Top: string; // You can remove this if not using Top property
13 }
14
15 const LOG_SOURCE: string = 'DynamicNavBarApplicationCustomizer';
16
17 export default class DynamicNavBarApplicationCustomizer extends BaseApplicationCustomizer<IDynamicNavBarApplicationCustomizerProperties> {
18   private _topPlaceholder: PlaceholderContent | undefined;
19
20   public onInit(): Promise<void> {
21     Log.info(LOG_SOURCE, 'Initialized DynamicNavBarApplicationCustomizer');
22
23     // Register event handler for placeholder changes
24     this.context.placeholderProvider.changedEvent.add(this, this._renderPlaceholders);
25     this._renderPlaceholders();
26
27     return Promise.resolve();
28   }
29
30   private _renderPlaceholders(): void {
31     // Render to the top placeholder
32     if (!this._topPlaceholder) {
33       this._topPlaceholder = this.context.placeholderProvider.tryCreateContent(
34         PlaceholderName.Top,
35         { onDispose: this._onDispose.bind(this) } // Binding onDispose method
36       );
37     }
38   }
39
40   private _onDispose(): void {
41     // Unmount React component on dispose
42     if (this._topPlaceholder && this._topPlaceholder.domElement) {
43       ReactDOM.unmountComponentAtNode(this._topPlaceholder.domElement);
44     }
45   }
46 }
```

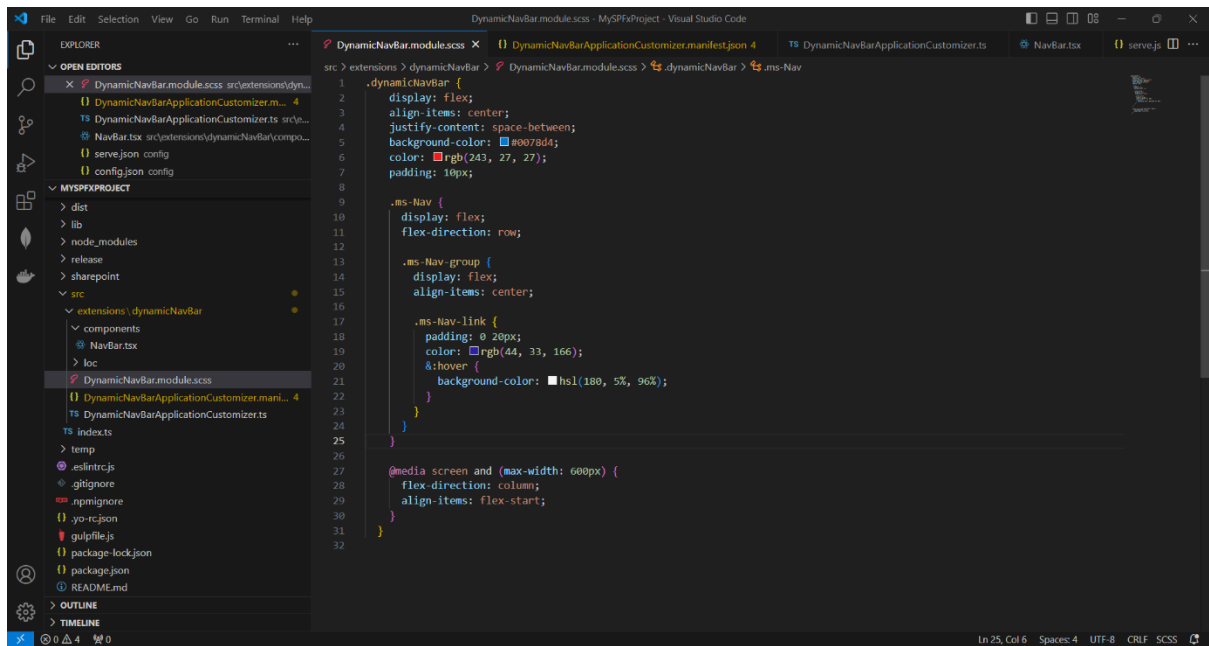


```
1 {
2   "$schema": "https://developer.microsoft.com/json-schemas/spfx/customizer/v1",
3   "id": "DynamicNavBarApplicationCustomizer",
4   "name": "Dynamic NavBar Application Customizer",
5   "version": "1.0.0",
6   "description": "Dynamic NavBar Application Customizer",
7   "category": "Navigation",
8   "author": "Microsoft",
9   "helpUrl": "https://aka.ms/NavBarCustomizer",
10   "images": {
11     "logo": "https://aka.ms/NavBarCustomizerLogo",
12     "tile": "https://aka.ms/NavBarCustomizerTile"
13   },
14   "components": [
15     {
16       "name": "DynamicNavBarApplicationCustomizer",
17       "type": "Component",
18       "url": "https://aka.ms/NavBarCustomizer"
19     }
20   ],
21   "properties": [
22     {
23       "name": "Top",
24       "type": "Text",
25       "defaultValue": "Top",
26       "description": "Top placeholder"
27     }
28   ],
29   "onDispose": "this._onDispose.bind(this)"
30 }
```

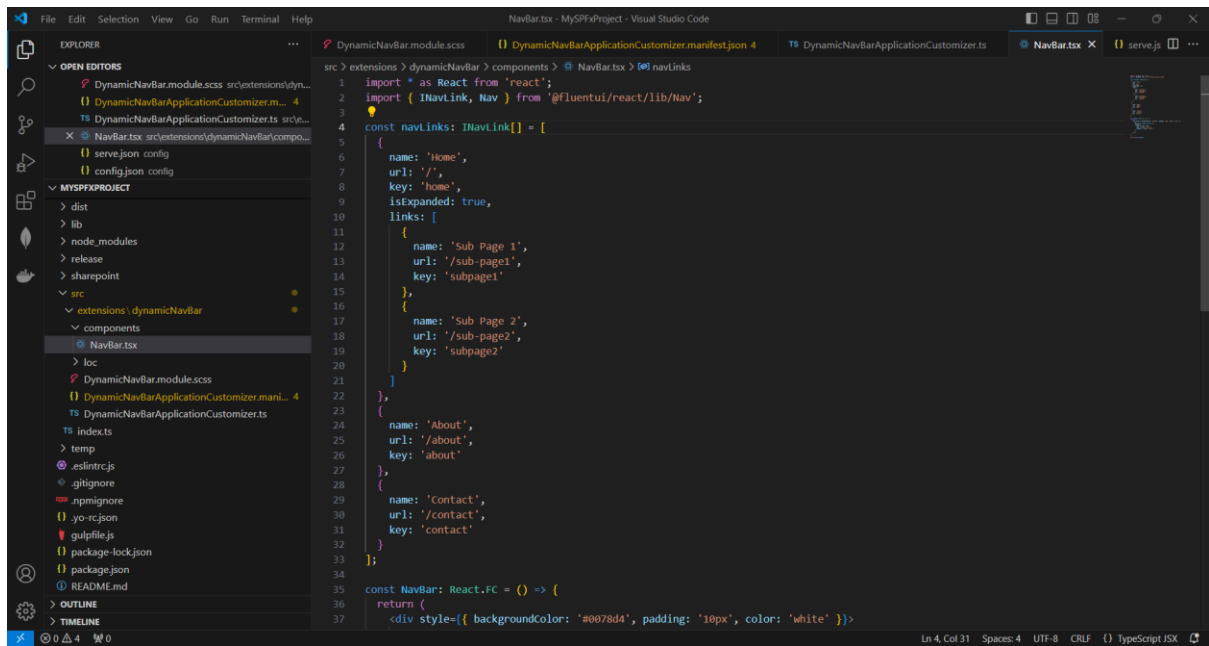
manifest.json file



.module.scss file



.tsx file



Output Screenshots

